

Problem

Use *PSpice* or *MultiSim* to verify the results in Example 5.9. Assume nonideal op amps LM324.

Example 5.9:

Find v_o and i_o in the circuit in Fig. 5.29.

Solution:

This circuit consists of two noninverting amplifiers cascaded. At the output of the first op amp,

$$v_a = \left(1 + \frac{12}{3}\right)(20) = 100 \text{ mV}$$

At the output of the second op amp,

$$v_o = \left(1 + \frac{10}{4}\right)v_a = (1 + 2.5)100 = 350 \text{ mV}$$

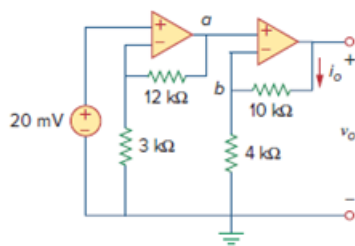
The required current i_o is the current through the 10-k Ω resistor.

$$i_o = \frac{v_o - v_b}{10} \text{ mA}$$

But $v_b = v_a = 100 \text{ mV}$. Hence,

$$i_o = \frac{(350 - 100) \times 10^{-3}}{10 \times 10^3} = 25 \text{ } \mu\text{A}$$

Figure. 5.29:



Step-by-step solution

Step 1 of 2

$$V_0 = A(V_2 - V_1) = 200(V_2 - V_1) \text{ .}$$

(A)

$$\begin{aligned} V_0 &= 200(0.386 - 0.402) \\ &= -3.2 \text{ V} \text{ .} \end{aligned}$$

Step 2 of 2

(B)

$$\begin{aligned} V_0 &= 200(1.011 - 1.002) \\ &= 1.8 \text{ V} \text{ .} \end{aligned}$$